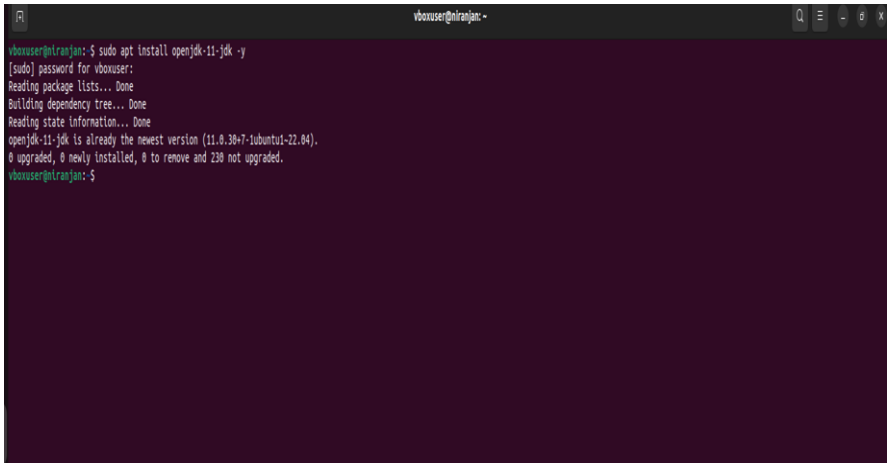
Program 1: Introduction to Maven: Overview of Build Automation Tools, Installation and Setup

Installation and Setup

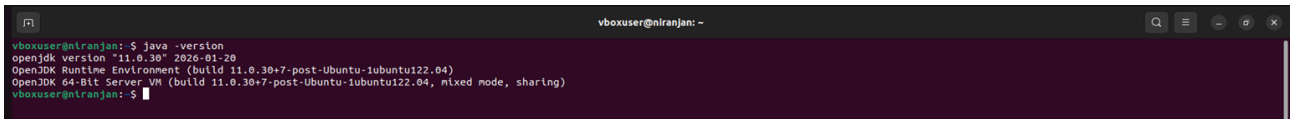
Java (JDK) Installation:

Step 1: Type the command **sudo apt install openjdk-11-jdk -y in the terminal**

It will install the OpenJDK 17 (Java Development Kit) and accepts installation automatically.



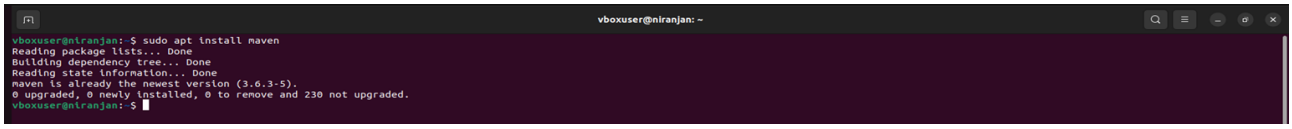
Step 2: Installation can be confirmed by running the command **java -version**



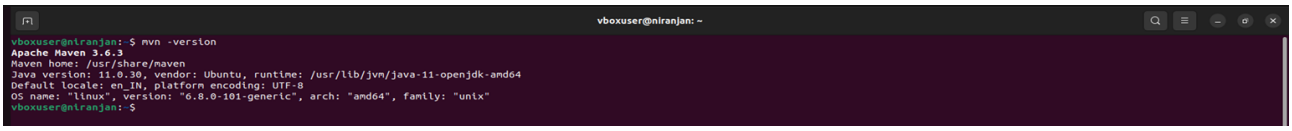
Maven Installation:

Step 1: Type the command **sudo apt install maven** in the terminal.

It will install Apache Maven from Ubuntu's package repository.



Step 2: Installation can be confirmed by running the command **mvn -version**



Gradle Installation:

Step 1: Type the command **sudo apt install gradle** in the terminal

It installs Gradle build automation tool from Ubuntu's package repository.

```
vboxuser@niranjan: ~
vboxuser@niranjan:~$ sudo apt install gradle
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gradle is already the newest version (4.4.1-13).
0 upgraded, 0 newly installed, 0 to remove and 230 not upgraded.
vboxuser@niranjan:~$
```

Step 2: To check the proper installation of Gradle run the command **<u>gradle –version</u>**

```
vboxuser@niranjan: ~
vboxuser@niranjan:~$ gradle -version
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.codehaus.groovy.reflection.CachedClass (file:/usr/share/java/groovy-all.jar) to method java.lang.Object.finalize()
WARNING: Please consider reporting this to the maintainers of org.codehaus.groovy.reflection.CachedClass
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release

------------------------------------------------------------
Gradle 4.4.1
------------------------------------------------------------

Build time:   2012-12-21 00:00:00 UTC
Revision:     none

Groovy:       2.4.21
Ant:          Apache Ant(TM) version 1.10.12 compiled on January 17 1970
JVM:          11.0.30 (Ubuntu 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OS:           Linux 6.8.0-101-generic amd64

vboxuser@niranjan:~$
```